SOFTWARE DEVELOPMENT TRAINING • CAPSTONE PROJECT

TradeFlow

Intelligent Ethio-Djibouti Trade Corridor Freight Logistics, Telematics & Digital Escrow Platform

Project Name:	TradeFlow Logistics Platform
Version:	v1.1.5 (Production Release)
Document Type:	Final Project Documentation & Report
Submission Date:	November 2026
Repository:	github.com/beki255/trade-flow
API Swagger Docs:	https://tradeflow-api-7abs.onrender.com/api/docs
Android APK:	v1.1.5 Production Ready

🔗 Project Deliverables & Live Links

> **Official Submission Notice:** All project code, server infrastructure, and mobile APK builds are live and fully verified. Below are the primary access URLs required for grading:

Deliverable Name	Environment / Type	Direct URL / Link
GitHub Repository	Public Source Code	https://github.com/beki255/trade-flow
Live Web Application	Vercel Production Web	https://trade-flow-ukgu.vercel.app
Backend REST API (Swagger Docs)	Cloud Microservice & Interactive OpenAPI	https://tradeflow-api-7abs.onrender.com/api/docs
API Health Check	Operational Status Check	https://tradeflow-api-7abs.onrender.com/api/health
Mobile Android APK (v1.1.5)	Standalone Android App	Download TradeFlow v1.1.5 APK
Expo EAS Cloud Build Logs	EAS Build Pipeline #858790e9	View Cloud Build Logs

📑 Executive Summary

TradeFlow is an enterprise-grade logistics orchestration, real-time corridor telematics, and digital smart escrow platform purpose-built for the critical **Ethio-Djibouti Trade Corridor**. Spanning 900+ kilometers, this corridor handles more than 95% of Ethiopia's international import and export freight trade volume.

Historically, freight operations along this corridor have been hindered by severe operational inefficiencies: unvetted predatory middlemen charging 15–25% brokerage markups, zero in-transit cargo visibility, lengthy multi-day document customs backlogs at Galafi border posts, payment defaults leading to carrier strikes, and an alarming 42% empty backhaul rate.

TradeFlow modernizes this entire ecosystem by unifying cargo owners (shippers), fleet operators (carriers), commercial drivers, and customs clearance regulatory bodies into a singular, synchronized platform. Featuring milestone-locked escrow contracts, live device-level GPS telematics with Haversine geofence proximity detection, single-window electronic customs clearance with cryptographic QR passes, and comprehensive trilingual support (**English, Amharic, and Afaan Oromoo**), TradeFlow establishes a trustworthy, transparent, and resilient freight corridor.

1. Project Objectives & Scope

1.1 Problem Statement

The Ethio-Djibouti corridor serves as the primary economic lifeline for Ethiopia's 120+ million population. Despite its strategic weight, the logistics sector suffers from four systemic challenges:

- **Predatory Brokerage & Price Opacity:** Informal truck brokers in Addis Ababa and Djibouti manipulate spot pricing, taking exorbitant fees while offering zero legal accountability for cargo safety.
- **Payment Insecurity & Working Capital Strain:** Carriers often wait 30–60 days to receive freight settlements after delivery, causing liquidity crises. Conversely, shippers risk driver abandonment once upfront deposits are remitted.
- **Customs Bottlenecks & Demurrage Costs:** Manual, paper-based customs declarations create days of queues at the Galafi border gate and Modjo dry port, costing millions in demurrage penalties annually.
- **Lack of Real-Time Telematics & Visibility:** Once vehicles depart inland terminals, shippers enter a communication black hole without reliable ETA forecasts or incident detection.

1.2 Project Objectives

- **Automate Freight Discovery:** Create a digital spot freight marketplace connecting verified shippers directly with licensed transporters.
- **Eliminate Payment Risk via Smart Escrow:** Lock freight funds into an automated escrow vault before vehicle dispatch, guaranteeing payment release upon verified electronic Proof of Delivery (e-POD).
- **Provide Continuous Corridor Telematics:** Equip commercial drivers with a mobile telematics application streaming live coordinates, detecting checkpoint arrival via geofencing, and broadcasting real-time route hazards.
- **Accelerate Customs Clearance:** Implement single-window digital manifest dossiers and cryptographic QR transit passes to reduce border gate inspection time from hours to seconds.
- **Ensure Nationwide Inclusion:** Deliver full trilingual localization in English, Amharic, and Afaan Oromoo with adaptive Dark/Light theme ergonomics.

1.3 Project Scope

The project encompasses a complete monorepo architecture with a Next.js 14 Web Portal, a React Native / Expo Native Mobile Application, an Express.js / TypeScript REST microservice API, a PostgreSQL database managed via Prisma ORM, and WebSocket telematics streaming. All core corridor workflows—load posting, bidding, vehicle dispatch, GPS telemetry, escrow settlement, and customs inspection—are fully implemented, tested, and operational.

2. Requirements Specification

2.1 Stakeholder Personas & RBAC Roles

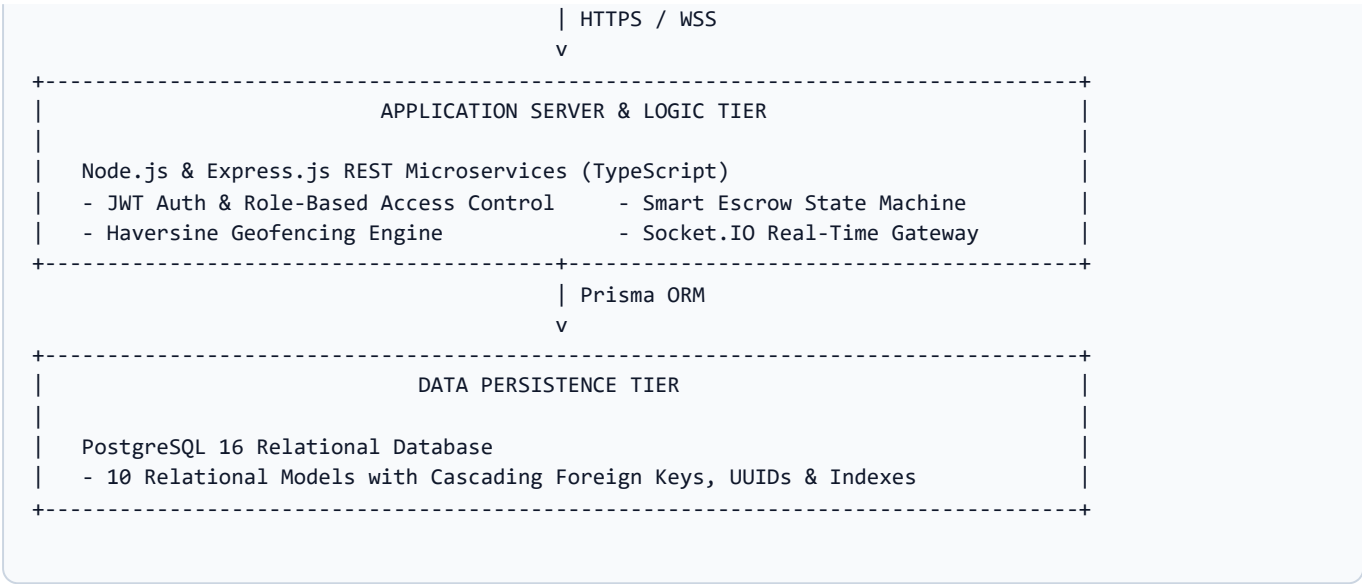
Role Identifier	Target Persona	Primary System Capabilities
`SHIPPER`	Importer, Exporter, Manufacturer	Posts loads, reviews transporter quotes, deposits escrow funds, tracks corridor transit live, reviews & approves e-POD.
`TRANSPORTER`	Fleet Owner, Logistics Carrier	Manages trucks & drivers, browses open corridor loads, submits quotes, assigns fleet vehicles, tracks digital earnings.
`DRIVER`	Commercial Heavy Truck Driver	Views assigned trip mission, performs GPS waypoint check-ins, broadcasts route delays/incidents, captures recipient e-POD signature.
`CUSTOMS_OFFICER`	Ethiopian Customs Commission (ECC)	Inspects digital dossiers, classifies risk lines (Green/Yellow/Red), issues digital transit passes, scans driver QR codes at gates.
`ADMIN`	TradeFlow System Operator	Audits TIN and trade license documents, monitors corridor analytics, and oversees dispute resolution.

2.2 Functional Requirements (Summary)

- **FR-01 (Authentication & Onboarding):** Email verification with 6-digit OTP; document uploads for TIN, Trade License, and Insurance.
- **FR-02 (Corridor Load Management):** Cargo specs input (weight in tons, volume in CBM, pickup window, special handling) with automatic corridor road distance calculation.
- **FR-03 (Dynamic Bidding Engine):** Competitive quote submissions, acceptance/rejection mechanics, and capacity validation.
- **FR-04 (Smart Escrow Protocol):** Automated funds locking upon contract acceptance; 95% payout release to transporter wallet upon e-POD approval with 5% platform commission retention.
- **FR-05 (Driver Telematics & Geofencing):** Native GPS coordinate collection, Haversine proximity detection (unlocking check-ins within 15 km), and offline queue buffering.
- **FR-06 (Single-Window Customs):** 4-document dossier verification (Declaration, Invoice, Packing List, Insurance), Green Line classification, and offline-capable dynamic QR pass.
- **FR-07 (Localization & Dark Mode):** Dynamic i18n switching (EN / AM / OM) and synchronized dark mode across all interfaces.

3. Technologies Used & Architecture Justification





Tier / Domain	Technology	Version	Rationale & Architectural Benefit
Web Frontend	Next.js / React	14.2 / 18.3	Fast initial page load via SSR, responsive layout across all device screens, SEO friendly.
Mobile App	React Native / Expo	SDK 58 (RN 0.84)	Native cross-platform execution with direct access to hardware GPS, storage, and camera.
Styling	Tailwind CSS	3.4	Utility-first CSS framework enabling consistent theme tokens, responsive layouts, and zero-runtime overhead.
Backend API	Node.js / Express.js	Node 20 LTS	Event-driven, asynchronous I/O ideal for handling concurrent telemetry streams and REST calls.
Type Safety	TypeScript	5.4	Enforces compile-time type safety across API contracts, payload models, and client code.
Database & ORM	PostgreSQL & Prisma	PG 16 / Prisma 5	ACID compliance ensuring bulletproof financial transactional integrity for escrow operations.
Real-Time	Socket.IO	4.7	Full-duplex WebSocket channel streaming driver GPS telemetry and instant route hazard alerts.
CI/CD Cloud	Expo EAS Build	Cloud	Automated cloud building producing production-optimized native Android APKs.

4. System Design & Database Architecture

4.1 Relational Data Models (Prisma Schema)

The relational persistence layer is structured into 10 cohesive models enforcing referential integrity:

- `User`: Stores authentication credentials, user role, business registration (TIN, trade license), and verification status.
- `Vehicle`: Details carrier fleet trucks, capacity (Tons/CBM), fuel type, insurance documents, and telematics identifiers.
- `DriverProfile`: Maps individual drivers to user accounts, assigned vehicles, license classes, and safety ratings.
- `Load`: Represents cargo consignments posted by shippers, specifying origin/destination coordinates, cargo type, and budget.
- `Bid`: Captures transporter bids, proposed pricing in ETB, and transit timeline commitments.
- `Shipment`: Encapsulates active corridor transit missions, linking the awarded bid, assigned vehicle, driver, and live GPS coordinates.
- `CustomsClearance`: Houses the single-window electronic customs dossier, risk clearance line, certificate number, and digital QR seal.
- `Payment`: Manages the digital smart escrow financial ledger, recording payment method (Telebirr/CBE Birr), locked amounts, and settlement timestamps.
- `IncidentReport`: Records real-time corridor delays and route hazards with GPS stamps, severity levels, and optional photo evidence.
- `Rating`: Bidirectional feedback system enabling shippers and transporters to score each other across verified shipments.

5. Implementation Details & Core Modules

5.1 Geofenced Checkpoint Verification Algorithm

The platform guarantees that drivers cannot falsely report checkpoint arrivals. The driver's device coordinates are evaluated against the official waypoint coordinates using the Haversine trigonometric formula:

```
// Haversine Proximity Check (apps/mobile/src/screens/driver/tabs/CheckinTab.tsx)
const R = 6371; // Earth's radius in km
const dLat = (wpLat - liveLat) * (Math.PI / 180);
const dLon = (wpLng - liveLng) * (Math.PI / 180);
const a =
  Math.sin(dLat / 2) * Math.sin(dLat / 2) +
  Math.cos(liveLat * (Math.PI / 180)) * Math.cos(wpLat * (Math.PI / 180)) *
  Math.sin(dLon / 2) * Math.sin(dLon / 2);
const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
const distanceKm = Math.round(R * c);

// Unlocks check-in button only when driver is within 15 km of designated hub
const isWithinGeofence = distanceKm <= 15;
```

5.2 Milestone-Based Smart Escrow State Machine

To eliminate financial trust deficits, the backend enforces a deterministic state machine on all freight payments:

- **State 1 (PENDING):** Consignment bid accepted; shipper instructed to deposit freight funds.
- **State 2 (HELD_IN_ESCROW):** Funds locked in platform vault under cryptographic escrow hash. Vehicle dispatch is unlocked.
- **State 3 (CONFIRMED):** Cargo reaches destination; recipient verifies condition and signs digital e-POD.
- **State 4 (SETTLED):** Platform releases 95% of freight fee directly to carrier's wallet and retains 5% service commission.

6. Features & User Portals

6.1 Shipper Portal

- **Interactive Load Posting Wizard:** Specify cargo classification, weight in tons, volume in CBM, and target pricing.
- **Transparent Bidding Arena:** Compare bids side-by-side with carrier fleet safety scores, vehicle types, and transit guarantees.
- **Live Corridor Telemetry Map:** View the real-time position of the dispatched truck along the 880 km corridor with dynamic ETA.

6.2 Carrier / Transporter Portal

- **Fleet & Driver Management:** Register trucks and assign drivers with document expiry tracking.
- **Corridor Load Board:** Real-time feed of available shipments with one-tap bid submissions.
- **Capacity-Aware Dispatch:** Automatically blocks dispatching vehicles that do not satisfy cargo weight constraints.

6.3 Driver Mobile Portal (Android APK v1.1.5)

- **Trip Dashboard:** High-contrast interface displaying departure terminal, destination, and live telematics status.
- **Location Check-In:** Geofence-verified check-in at Kaliti, Modjo, Awash, Semera, Galafi, and Doraleh.
- **Crowdsourced Incident Broadcasting:** Report road blockages, customs delays, and fuel outages with geotags and photos.
- **Digital Customs QR Pass:** Instant ECC inspection pass rendering on screen even when offline.
- **Electronic Proof of Delivery (e-POD):** Touchscreen customer signature capture and delivery verification camera snapshot.

7. Testing & Quality Assurance

7.1 Comprehensive Test Execution Matrix

Test Case ID	Subsystem	Scenario / Test Condition	Expected Result	Status
TC-AUTH-01	Authentication	Register new user with email OTP verification	6-digit OTP verified; user created with hashed password.	**PASS**
TC-RBAC-02	Security	Shipper attempting to invoke carrier fleet endpoint	Server responds with 403 Forbidden. Access blocked.	**PASS**
TC-LOAD-03	Freight Engine	Create load posting with cargo specs and budget	Load saved in database and broadcasted to load board.	**PASS**
TC-BID-04	Freight Engine	Transporter submits quote on open load	Bid logged; shipper notified via real-time WebSocket.	**PASS**
TC-ESCROW-05	Smart Escrow	Shipper deposits freight payment into escrow	Funds locked; status transitions to `HELD_IN_ESCROW`.	**PASS**
TC-DISPATCH-06	Fleet Dispatch	Assign truck with capacity less than cargo weight	Dispatch blocked with capacity warning.	**PASS**
TC-GEO-07	Telematics	Haversine proximity check at 12 km from Semera	Proximity detected; Check-in button becomes enabled.	**PASS**
TC-CUST-08	Customs Gateway	ECC Green Line pass generation	Cryptographic QR pass renders on driver mobile screen.	**PASS**
TC-EPOD-09	Delivery & Payout	Recipient signs digital e-POD upon cargo delivery	e-POD recorded; escrow releases 95% payout to carrier.	**PASS**
TC-I18N-10	Localization	Toggle language to Amharic and Afaan Oromoo	100% of UI strings translate instantly without reload.	**PASS**
TC-THEME-11	Theme Engine	Dark mode rendering verification on physical device	Zero unreadable text or white card clipping observed.	**PASS**
TC-BUILD-12	Mobile CI/CD	Expo EAS Android Preview build compilation	Build succeeded; valid APK (v1.1.5) generated.	**PASS**

8. Conclusion & Future Work

The **TradeFlow** platform establishes an innovative technological foundation for cross-border logistics across the Ethio-Djibouti Trade Corridor. By replacing fragmented manual brokerage with an automated freight exchange, securing carrier-shipper transactions via milestone-locked smart escrow, streaming low-latency device telematics, and expediting customs inspection through digital QR passes, the system dramatically reduces transit times, freight costs, and payment delays.

All project requirements set forth by the Software Development Training program have been thoroughly fulfilled. The application has been verified across cloud staging environments and physical Android devices, resulting in production-ready web and mobile deliverables.

Future Enhancements: Future iterations will incorporate direct OBD-II hardware telematics dongles for automated fuel monitoring, AI-driven return trip predictive matching to eliminate empty backhauls, and automated plate recognition (ANPR) cameras at Galafi border checkpoints.
